Efficiency Corrections Comparison: Current vs. Previous Results

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1 Methodology: Efficiency and Uncertainty Calculations

This section details the updated statistical methodologies used to evaluate the average efficiencies and propagate their associated uncertainties within each kinematic mass and x_F bin.

1.1 Reconstruction Efficiency and Correlated Uncertainties

The reconstruction efficiency $E_{\text{reco},i}$ for a single event i is interpolated from a global tracking efficiency curve evaluated at the chamber occupancy parameter, D_1. Because multiple events within the same kinematic bin frequently fall between the same D_1 interpolation nodes, their interpolated uncertainties σ_i are mathematically correlated.

To account for this, we construct an $N \times N$ covariance matrix for the N events within a given kinematic bin:

$$Cov_{i,j} = \rho_{i,j}\sigma_i\sigma_j \tag{1}$$

where the correlation coefficient $\rho_{i,j}$ is determined by the shared interpolation nodes:

- $\rho_{i,j} = 1$ if event i and event j share both bounding interpolation nodes (same D_1 bin).
- $\rho_{i,j} = 1$ if event i and event j share exactly one bounding interpolation node (adjacent D_1 bins).
- $\rho_{i,j} = 0$ otherwise (assumed strictly independent).

The mean reconstruction efficiency for the kinematic bin is evaluated as the arithmetic mean of the event-by-event efficiencies:

$$\bar{E}_{\text{reco}} = \frac{1}{N}\sum_{i=1}^{N} E_{\text{reco},i} \tag{2}$$

By accounting for the off-diagonal covariance elements, the correctly propagated, correlated uncertainty on the mean efficiency is:

$$\sigma_{\bar{E}_{\text{reco}}} = \frac{1}{N}\sqrt{\sum_{i=1}^{N}\sum_{j=1}^{N} Cov_{i,j}} \tag{3}$$

1.2 Final Efficiency Evaluation

The mean hodoscope efficiency $\bar{E}_{\text{hodo}}$ is derived similarly. However, because its event-by-event evaluations are independent of the global tracking D_1 curve, its uncertainty is evaluated assuming strictly independent events (a purely diagonal covariance matrix):

$$\sigma_{\bar{E}_{\text{hodo}}} = \frac{1}{N}\sqrt{\sum_{i=1}^{N} \sigma^2_{\text{hodo},i}} \tag{4}$$

Previously, the final efficiency was calculated by multiplying $E_{\text{reco}} \times E_{\text{hodo}}$ on an event-by-event basis prior to averaging. To properly isolate and utilize the independent error characteristics of the tracking and hodoscope responses, the current methodology calculates the final efficiency as the product of the independently evaluated bin averages:

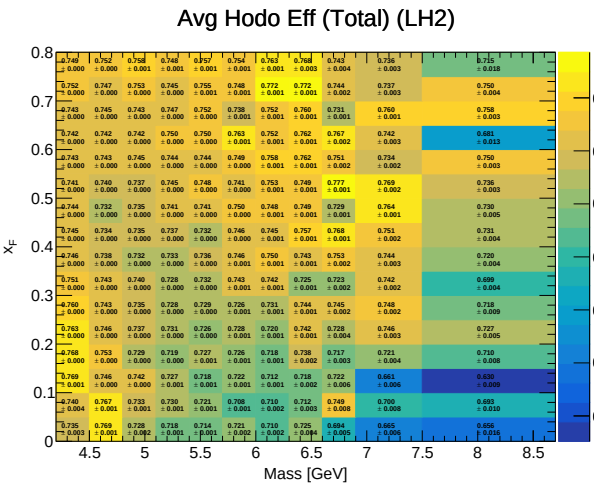
$$\bar{E}_{\text{final}} = \bar{E}_{\text{reco}} \times \bar{E}_{\text{hodo}} \tag{5}$$

The final propagated uncertainty on this product is evaluated using standard error propagation for independent variables:

$$\sigma_{\bar{E}_{\text{final}}} = \sqrt{\left(\bar{E}_{\text{hodo}} \cdot \sigma_{\bar{E}_{\text{reco}}}\right)^2 + \left(\bar{E}_{\text{reco}} \cdot \sigma_{\bar{E}_{\text{hodo}}}\right)^2} \tag{6}$$

2 LH2 Efficiency Corrections

2.1 Total Yield



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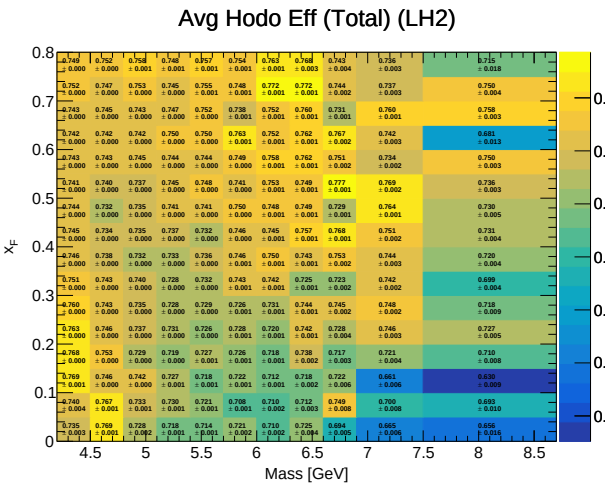
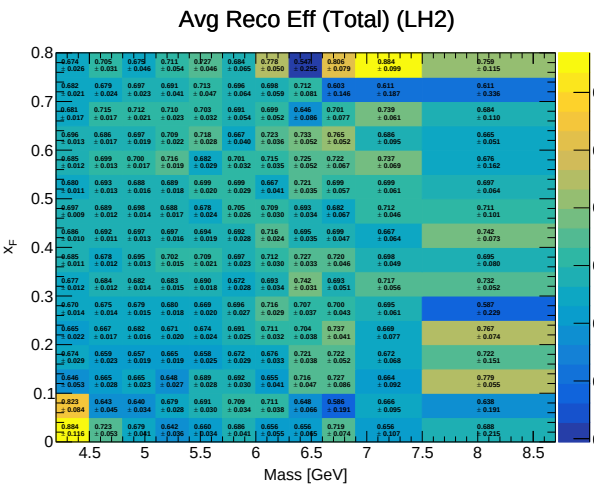


(b) Current

Figure 1: LH2 Total - Hodoscope Efficiency Corrections



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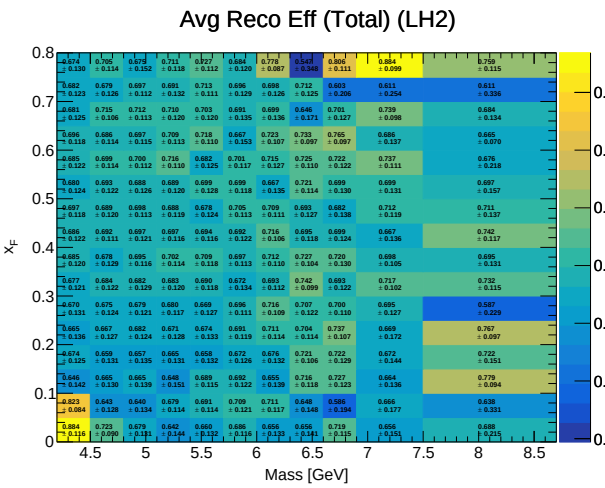

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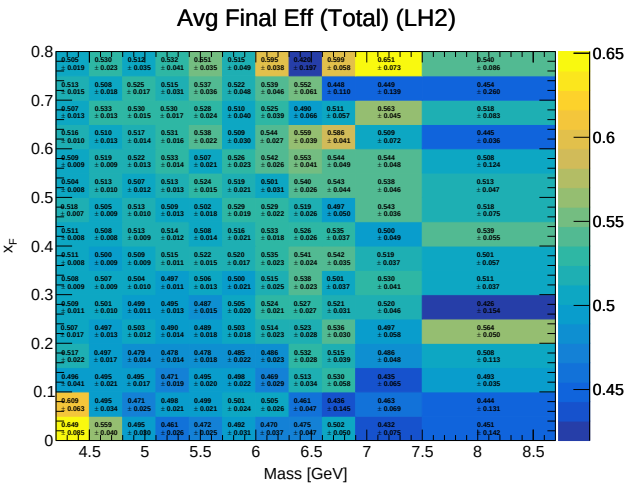
Figure 2: LH2 Total - Reconstruction Efficiency Corrections



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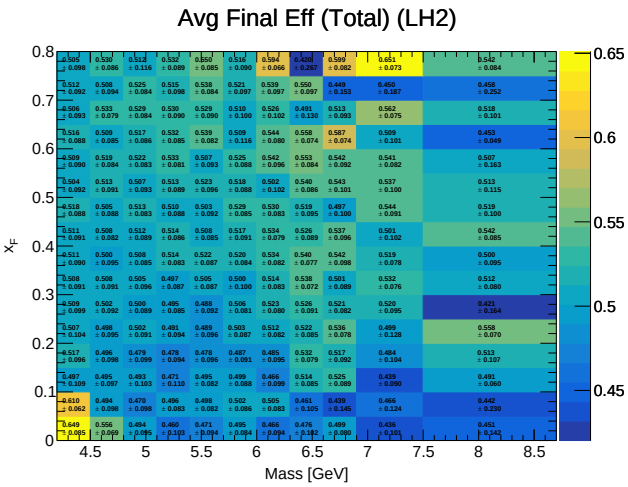
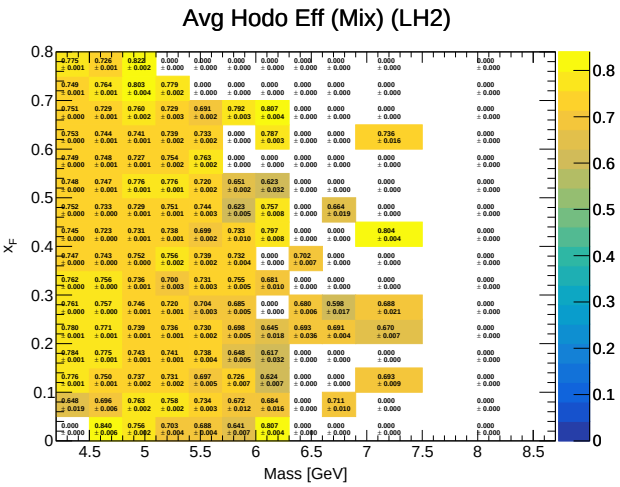


(b) Current

Figure 3: LH2 Total - Final Efficiency Corrections

2.2 Mix Yield



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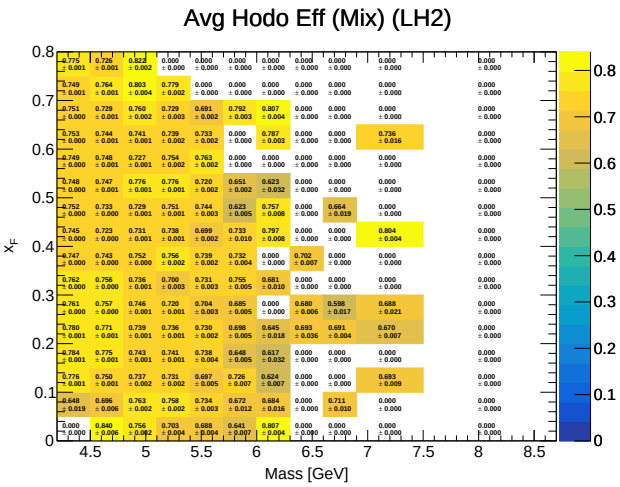
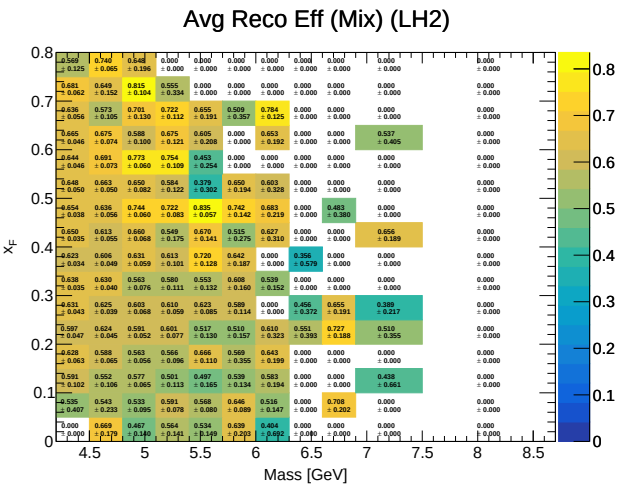


(b) Current

Figure 4: LH2 Mix - Hodoscope Efficiency Corrections



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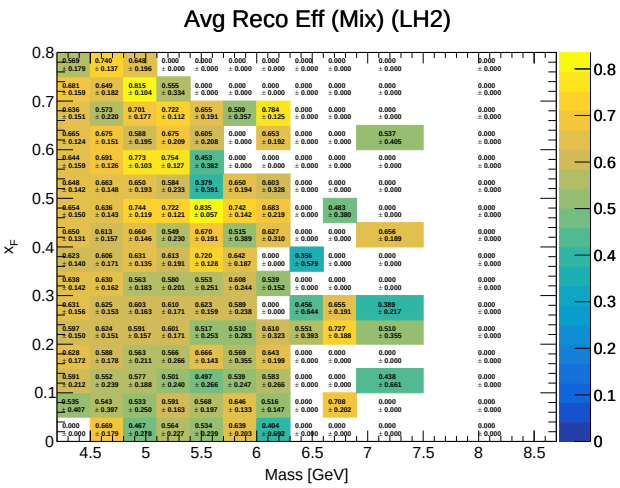

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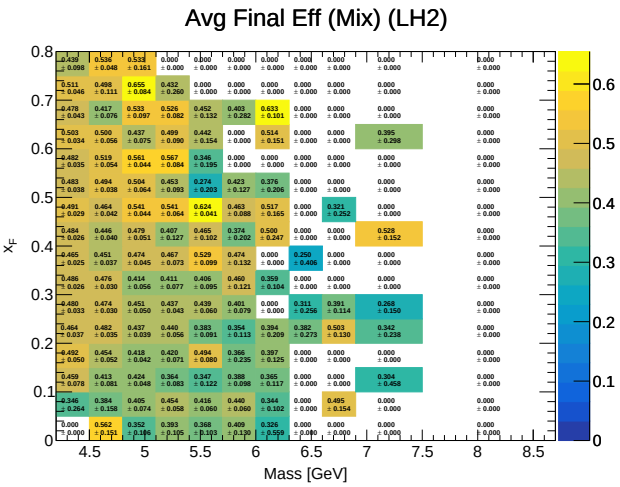
Figure 5: LH2 Mix - Reconstruction Efficiency Corrections



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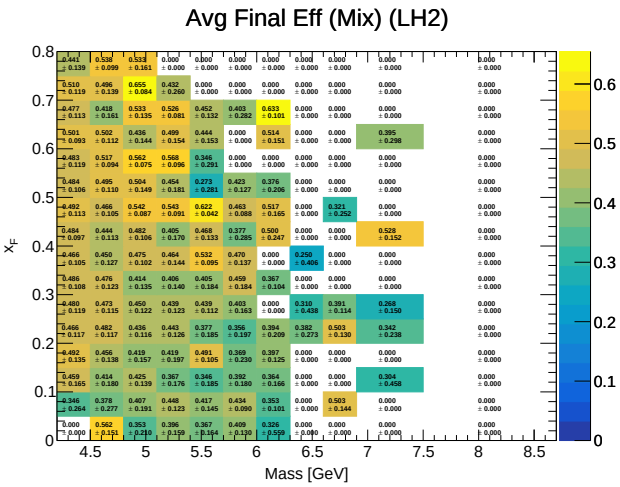
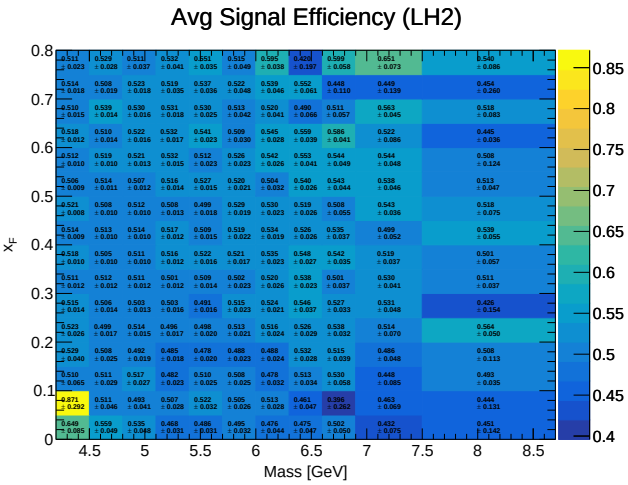


(b) Current

Figure 6: LH2 Mix - Final Efficiency Corrections

2.3 Signal Efficiency Corrections



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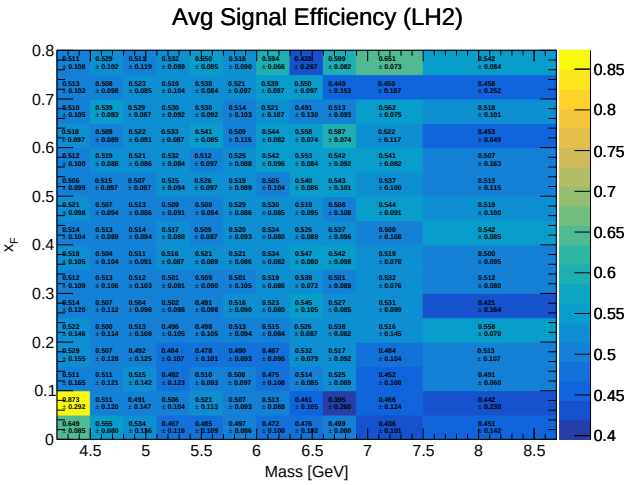

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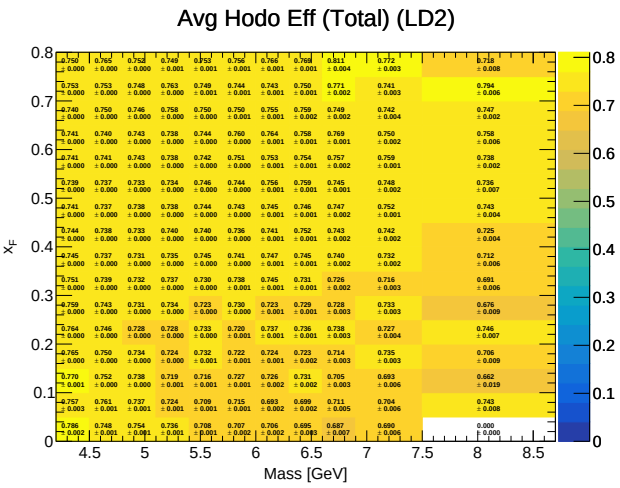
Figure 7: LH2 - Signal Efficiency Corrections

3 LD2 Efficiency Corrections

3.1 Total Yield



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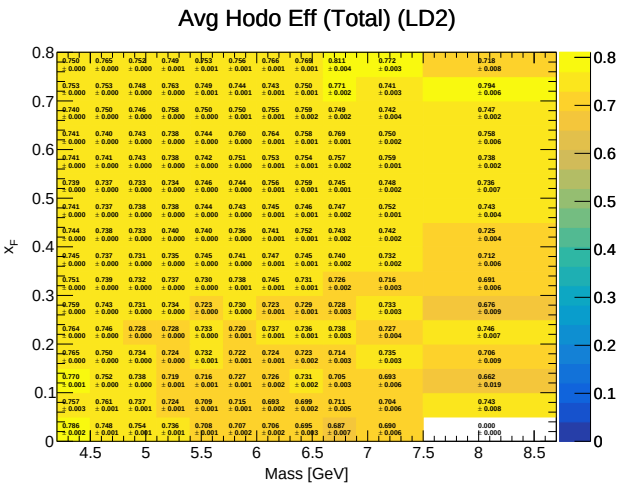

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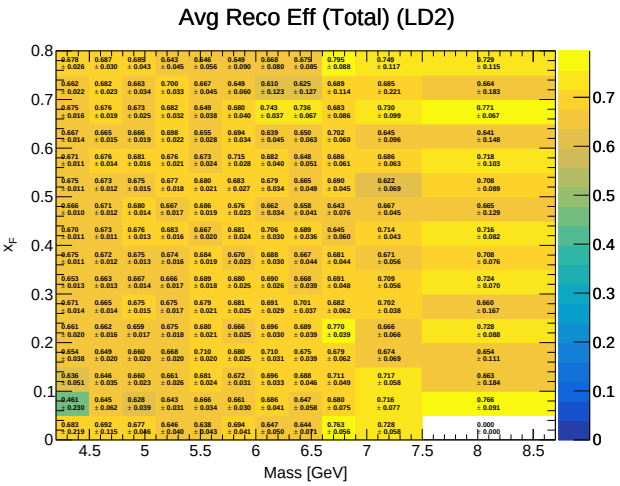
Figure 8: LD2 Total - Hodoscope Efficiency Corrections



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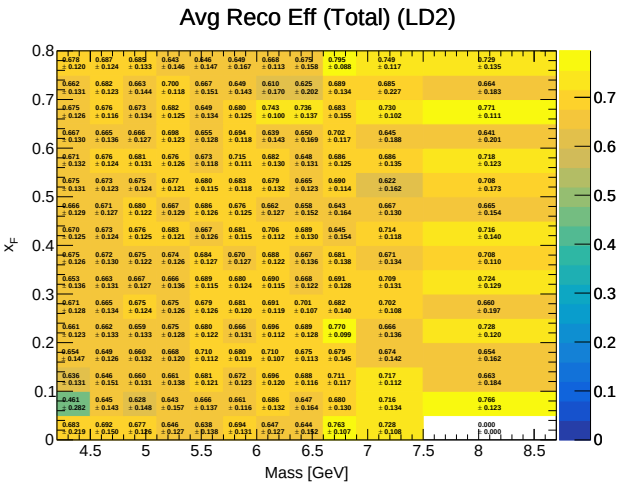

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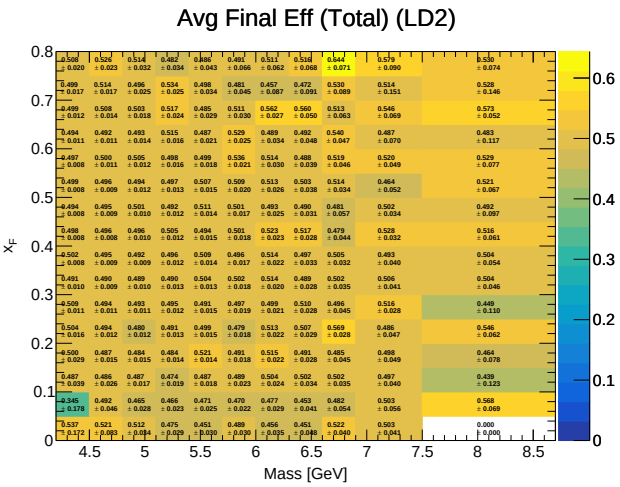
Figure 9: LD2 Total - Reconstruction Efficiency Corrections



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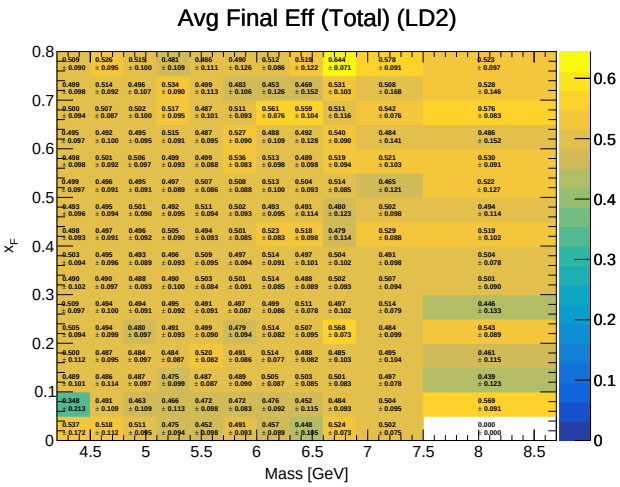

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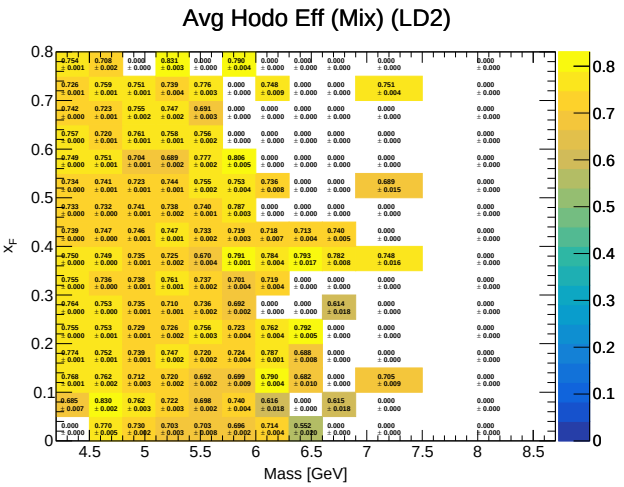
Figure 10: LD2 Total - Final Efficiency Corrections

3.2 Mix Yield



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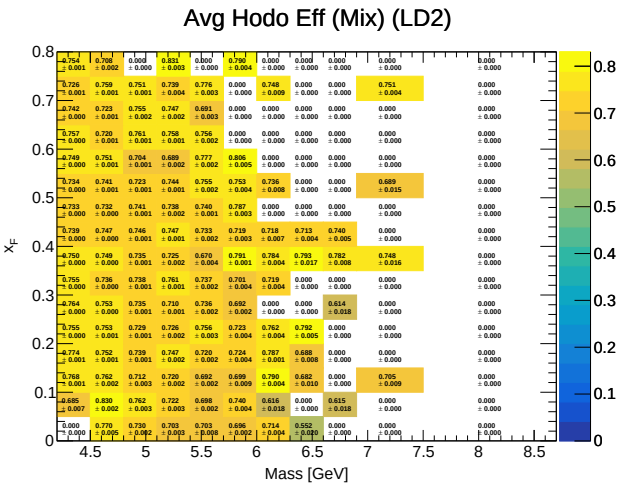

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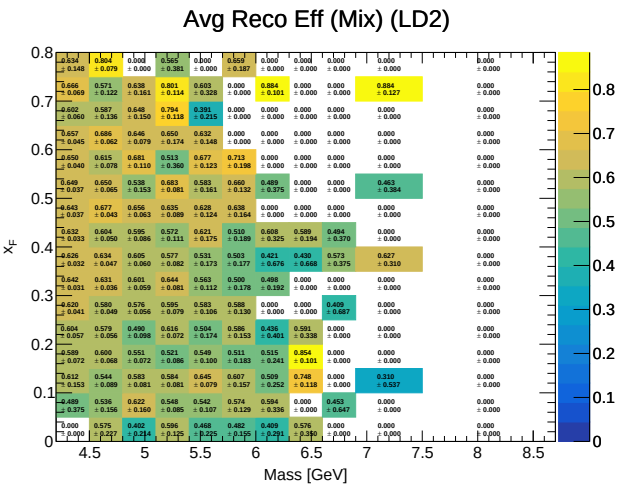
Figure 11: LD2 Mix - Hodoscope Efficiency Corrections



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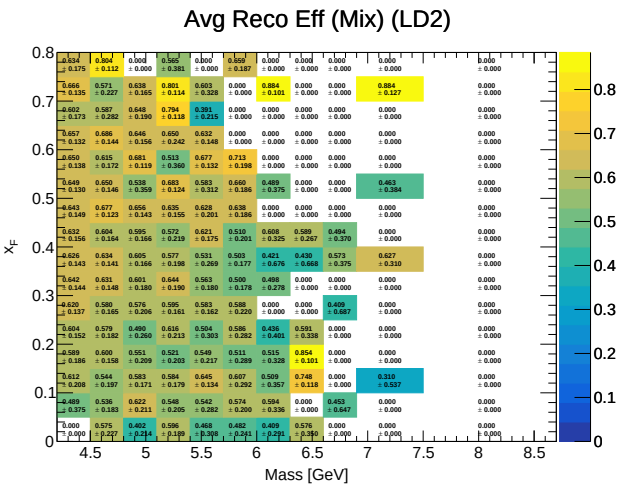

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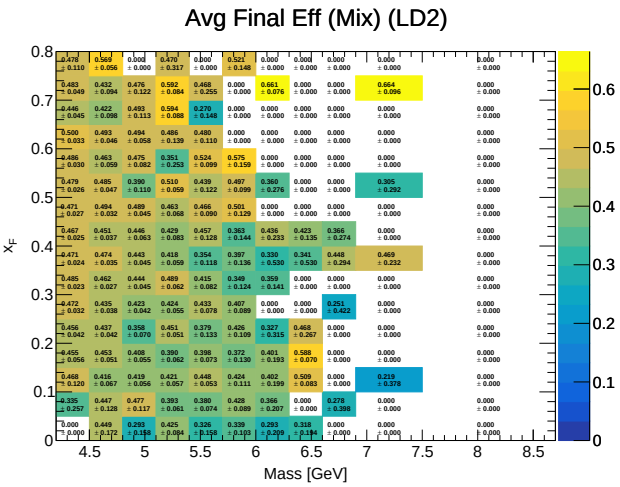
Figure 12: LD2 Mix - Reconstruction Efficiency Corrections



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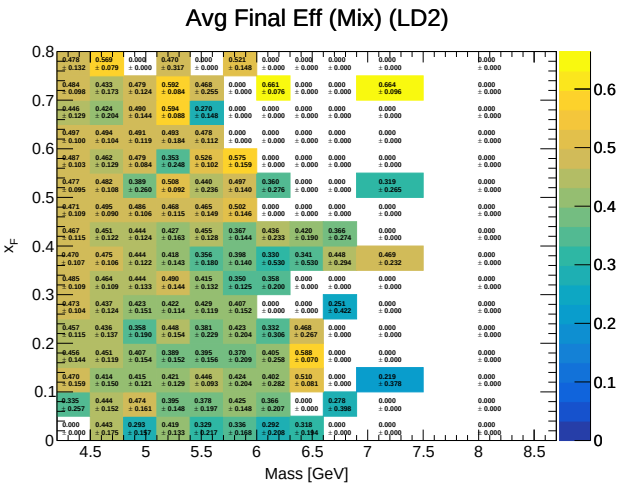

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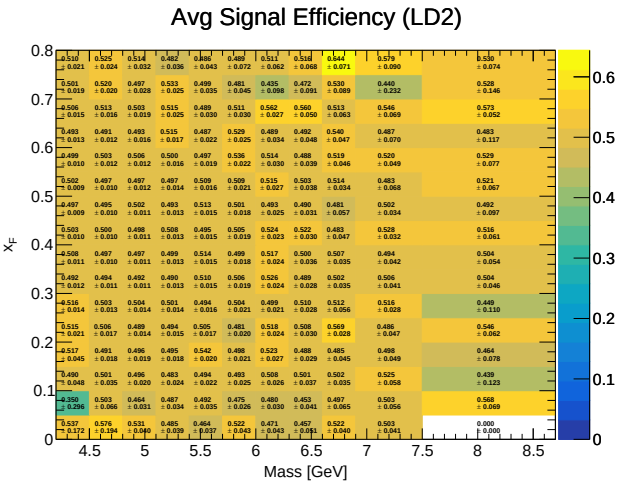
Figure 13: LD2 Mix - Final Efficiency Corrections

3.3 Signal Efficiency Corrections



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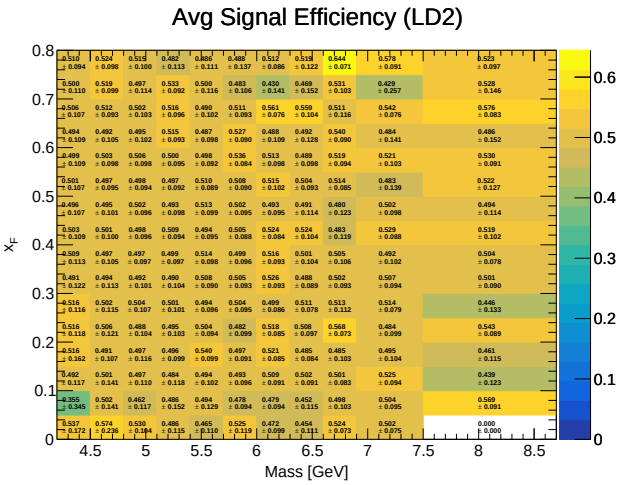

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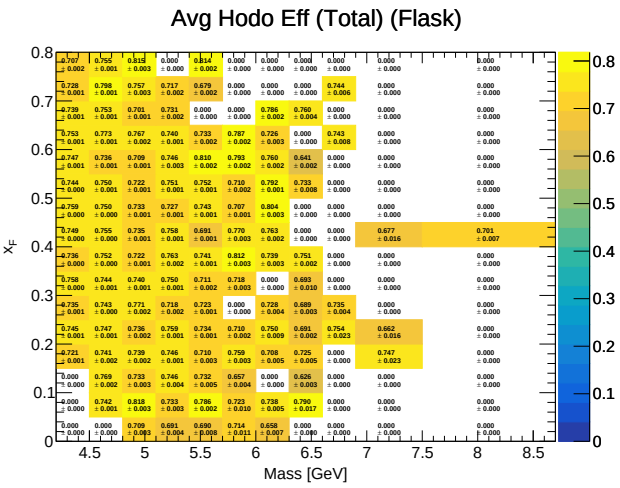
Figure 14: LD2 - Signal Efficiency Corrections

4 Flask Efficiency Corrections

4.1 Total Yield



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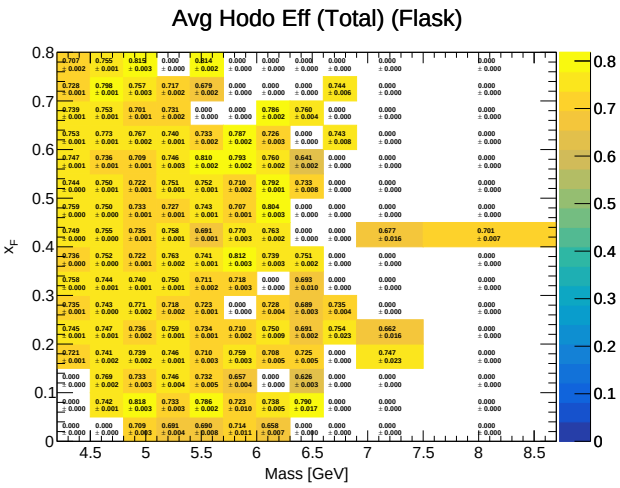
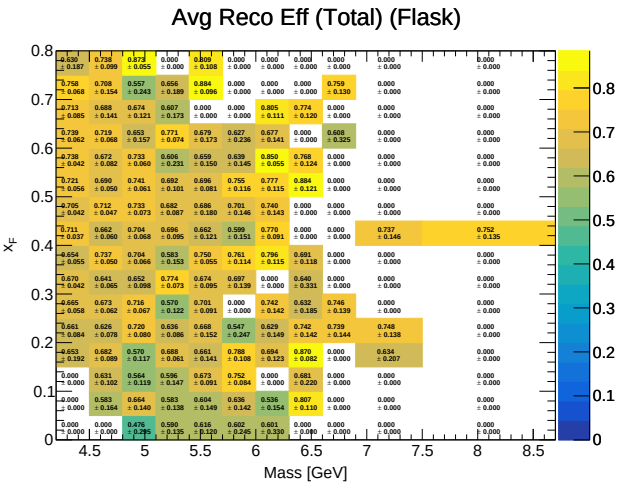


(b) Current

Figure 15: Flask Total - Hodoscope Efficiency Corrections



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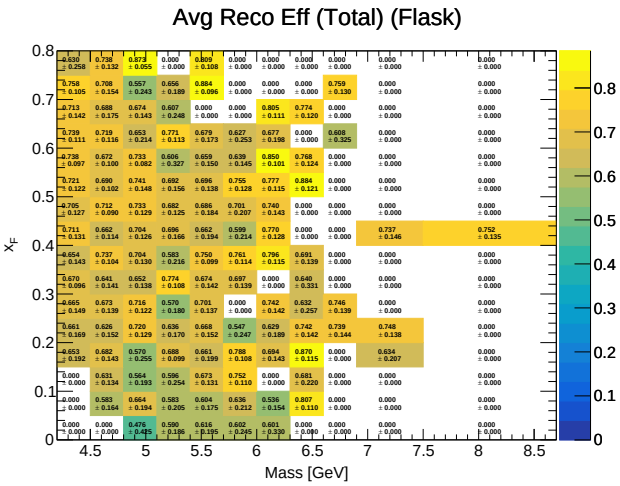
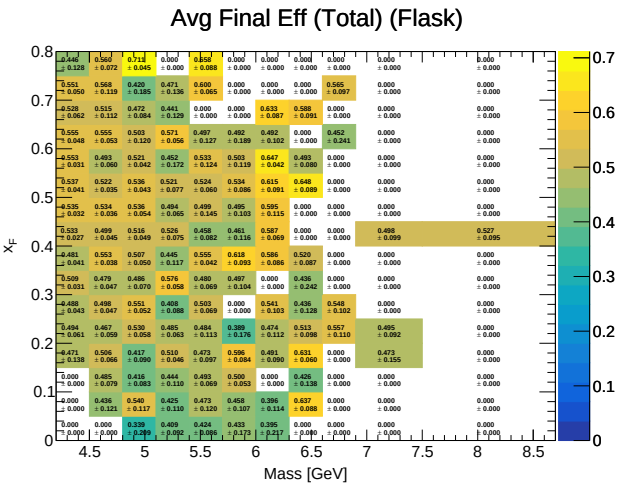


(b) Current

Figure 16: Flask Total - Reconstruction Efficiency Corrections



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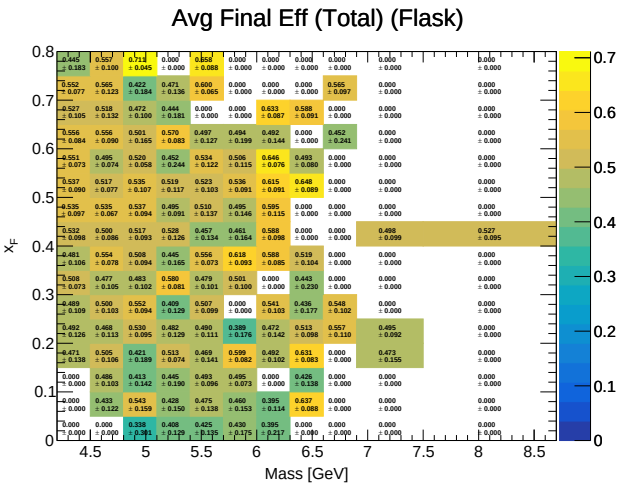

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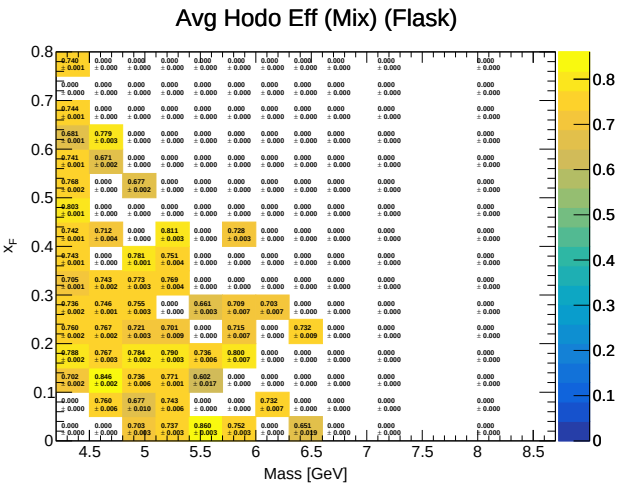
Figure 17: Flask Total - Final Efficiency Corrections

4.2 Mix Yield



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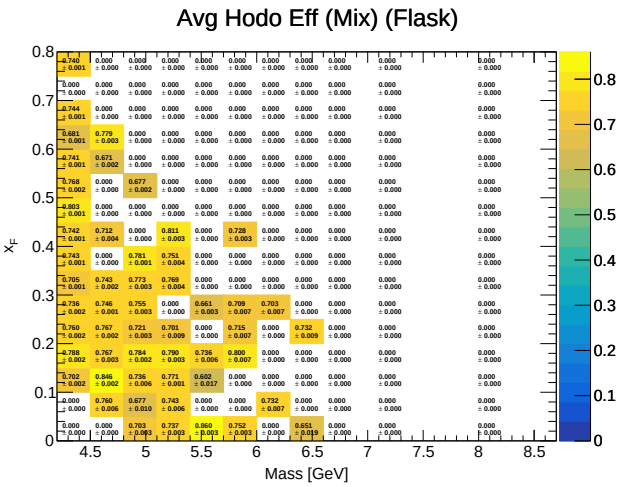
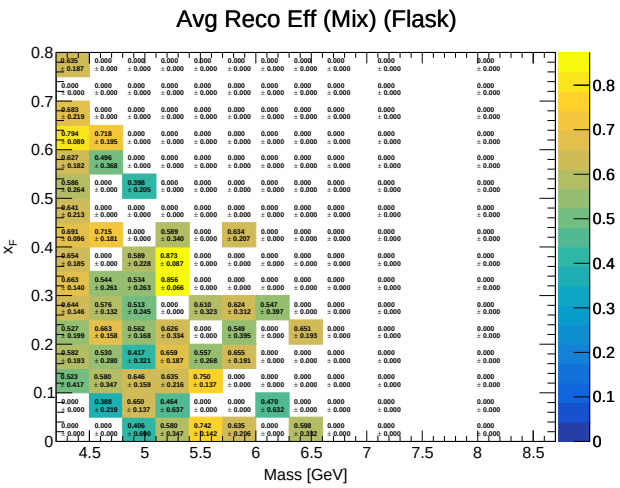


(b) Current

Figure 18: Flask Mix - Hodoscope Efficiency Corrections



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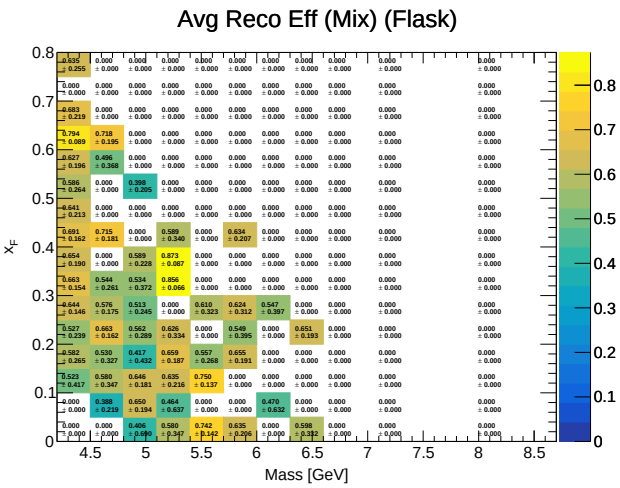
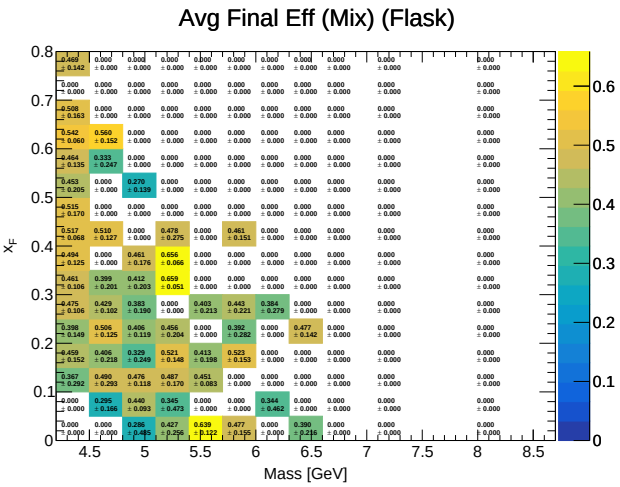


(b) Current

Figure 19: Flask Mix - Reconstruction Efficiency Corrections



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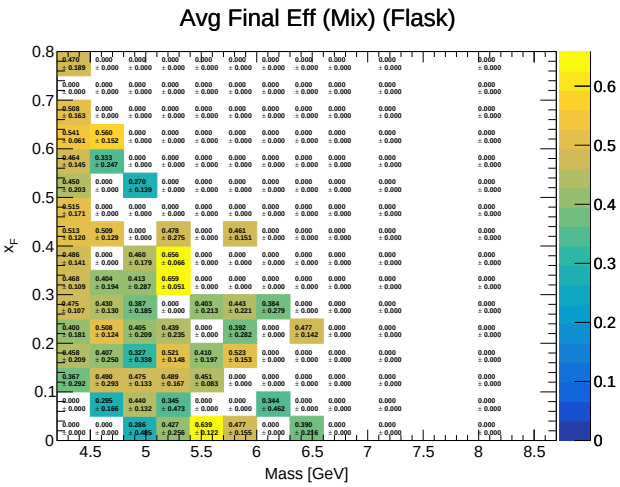

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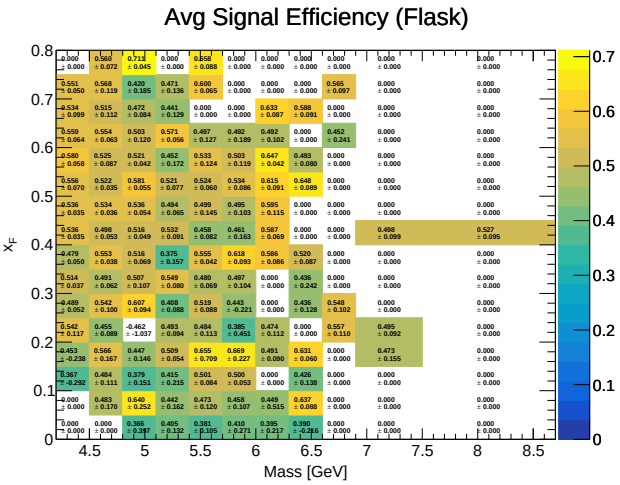
Figure 20: Flask Mix - Final Efficiency Corrections

4.3 Signal Efficiency Corrections



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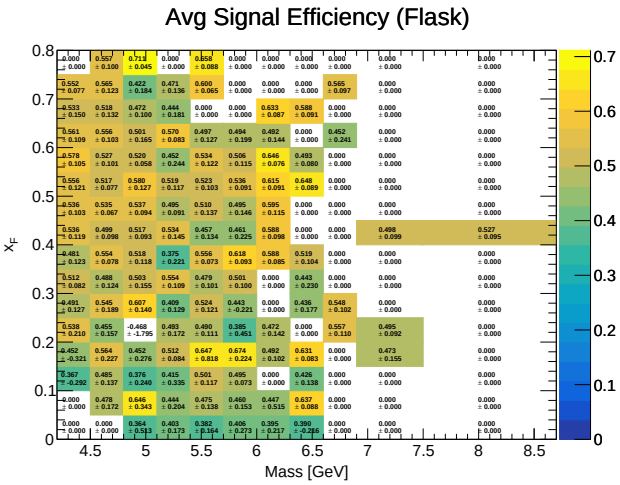

(b) Current

Figure 21: Flask - Signal Efficiency Corrections